<> CODEMETRIX

CODEMETRIX
From Coding Activity to Actionable Performance Insights

TEAM NAME CodeMetrix
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SCAN TO VIEW

CODEMETRIX ■

C O D E M E T R I X 01

THE CHALLENGE

Problem Statement
Students practice coding on platforms like LeetCode and GitHub, but faculty often lack a centralized system to track and evaluate that progress effectively.

■ Manual Tracking
Manual tracking of students' coding progress across individual
profiles.

■ Hard to Monitor at Scale
Difficult to monitor hundreds of students and keep records
continually updated.

■✕ Inactive Students Missed
Faculty cannot easily identify inactive students early for timely
guidance.

■ Scattered Data & Copy-Paste
Coding submissions are scattered across platforms without code authenticity verification.

■ No Easy Comparison
Difficult to compare section-wise performance, daily streaks, and authentic logic building.

■■ Database Storage Limits
Saving raw code files rapidly consumes free cloud quotas without an automated 50-limit purge.

C O D E M E T R I X 02

THE APPROACH

Proposed Solution
CodeMetrix is a centralized dashboard that automatically collects and analyzes students' LeetCode and GitHub activity, and presents it through student, section, and faculty dashboards.

■
LeetCode & GitHub

■
Automated Data Collection

■■

Data Processing & Forensics

■■

Supabase Database

CODEMETRIX

■ Student Dashboard ■ Faculty Analytics ■■ Admin Dashboard

C O D E M E T R I X 03

UNDER THE HOOD

Technology Stack

FRONTEND AUTOMATION & PROCESSING DATABASE & AUTHENTICATION

HTML
Builds the structure of the web dashboard

CSS
Creates the responsive and modern UI

JavaScript & JSZip
Handles interactions, filtering, search & 1-click ZIP downloads

Python
Automates LeetCode/GitHub data collection and processing

GitHub Actions
Runs automated trackers periodically without manual execution

Monaco DOM Hook
Browser extension capturing typing rhythm and paste counts

SQL & Triggers
Auto-replace deduplication constraints & 50-problem purge

Supabase
Stores student profiles, coding data and application data

Row Level Security
Guarantees secure access across faculty and student portals

C O D E M E T R I X 04

HOW IT WORKS

System Architecture

LeetCode
• Problems solved (E / M / H) • Daily activity & submissions

GitHub
• Repositories & commits • Contribution activity

Python — Processing Engine
• Fetch data • Process and clean data • Validate deduplication • Enforce 50-limit window

GitHub Actions
• Scheduled execution • Runs trackers automatically

Supabase (PostgreSQL)
• Central data store • Auto-prune triggers • Powers all dashboards

Student Dashboard
• Solved breakdown • 1-Click ZIP archive

Faculty Analytics
• Section comparison • In-modal code viewer

Admin Dashboard
• Anti-paste forensics • Excel reports

C O D E M E T R I X 05

WHAT IT DELIVERS

Key Features

■ Automated LeetCode & GitHub Tracking
Daily background scraping captures solved counts, streaks, and
difficulty breakdowns automatically.

■ Section-wise Analysis
Instant filtering across sections with ranked leaderboards and
detailed student modals.

■✕ Inactive Students Detection

Early identification of students with broken practice streaks for
prompt academic mentoring.

■■ Suspicious Solving & Anti-Paste
Extension tracks typing vs paste deltas to flag AI-pasted code while keeping short one-liners clean.

■ 50-Problem Download & Auto-Purge

1-click ZIP and Excel archive of solved files with automatic
database reset to maintain free cloud limits.

■ Duplicate Prevention
Re-submitted problems automatically replace older records,
guaranteeing zero duplicate entries.

C O D E M E T R I X 06

WHAT SETS US APART

Innovation

■ Unified Coding Intelligence
Combines LeetCode + GitHub activity into one intelligence layer.

■ Real-Time Extension Telemetry
Monitors Monaco editor keystroke rhythm directly in-browser with zero external server latency.

■■ Pattern-Based Integrity Checks
Physical keystroke counting and smart one-liner tolerance (<=220 chars) eliminate false alarms.

■■ Self-Pruning Cloud Architecture
Maintains perpetual free-tier operation via automated rolling 50-problem pruning and offline archives.

■ Data-Driven Reporting
Historical and section-wise data-driven reporting formatted for reviews.

✉■ Automated Delivery
Automatic report sending and email updates for class advisors.

C O D E M E T R I X 07

WHY IT MATTERS

Impact

IMPACT RESULT
Consistent Practice Daily coding habit fostered across all students
Reduced Workload Automated monitoring saves 15+ faculty hours weekly
Early Intervention Inactive students identified quickly for counseling
Data-Driven Decisions Meaningful analytics for section-level performance reviews
Academic Integrity Unusual solving patterns and copy-paste dumps flagged
Storage Sustainability 50-problem auto-pruning enables perpetual zero-cost operation

C O D E M E T R I X 08

WHAT'S NEXT

Future Scope

■ AI-Based Performance & Complexity Optimization Hints
■ Advanced Anomaly & Code Quality Detection
■ Complete Online Department Coding Test Platform
■ Mobile App & Automated Telegram/WhatsApp Notifications
■ Integration with Multiple Coding Platforms (Codeforces, HackerRank, GFG)

"The future vision of CodeMetrix is to evolve from a coding activity tracker into an intelligent, personalized coding education platform."

C O D E M E T R I X 09

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THANK YOU
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